

Processing Indian Languages .

The English Language is the form Subject - Verb - Object (SVO)
where as the Hindi/Kannada etc is of the form
Subject - Object - Verb (SOV)

Ex:- राम स्कूल गया गये गयी गये
 (S) (O) (V) (S) (O) (V)

Ram went to School
(S) (V) (O)

* Instead of prepositions we have post-position case markers

Ex:- बच्चेಗಳು रीचीबक्स में ಇದೆ.

Ex:- Dress is in the cupboard.

* Changes in the position of the words do not generally change the meaning.

Ex: ① ನನಗೆ ಪುಸ್ತಕ ಇದೆ ಇಷ್ಟೆ

② ಪುಸ್ತಕ ಇದೆ ನನಗೆ ಇಷ್ಟೆ.

* Indian languages use Verb complexes consisting of sequence of Verbs.

Ex: गा रहा है (ga raha hai) — Singing

 खेल रहा है (Khel raha hai) — playing

Paninian Framework.

Paninian Grammar was written by Panini in 500 BC in Sanskrit (original text being titled Astadhyayi) the framework can be used for other Indian Language and possibly some Asian Language as well.

Some Important features:

Indian Languages has the flexibility to allow words to change the position of Subject and Object.

(a) Maa bachhe ko khana detii hai
Mother gives food to the child.

(b) Bachhe ko maa khana detii hai
Mother gives food to the child.

In Indian Lang Some verbs eg: give, take also combine with other verbs to change the aspect and modality of the verbs.

Ex: Usne khana khayya
He ate food

Vaha chala
He moved

In Indian Languages, the nouns are followed by Post positions Instead of prepositions.

Ex: Rekha ke pitha
Father of Rekha.

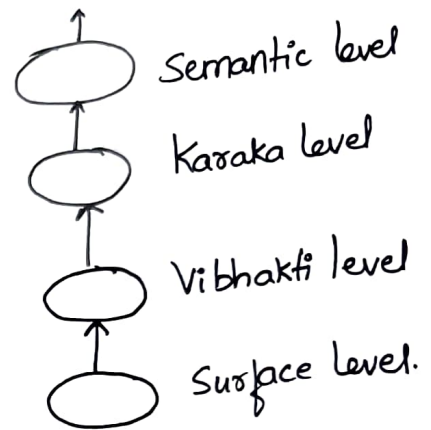
Layered Representation in PG:

The Language can be represented as →

1) Semantic Level:

This is where the speaker's intended meaning is captured without considering grammatical structure.

Ex: Ram read the book.



2) Karaka level:

The Semantic Roles of the sentence elements are identified such as agent, patient, instrument etc based on their relationship with the verb.

Ex: Ram (Karta) (agent)
Kitab (Karma) (patient)

3) Vibhakti level :- Ram takes the prathama Vibhakti and Kitab takes the Dvitiya Vibhakti this layer adds the grammatical case markers like nominative & Accusative.

4) Surface level: Observable form of the sentence,
Including word order and morphology

Ex: Ram ne kitab padhi